Definition

Let $f = \langle \mathbf{U} \mathbin{⨾} \mathbf{W}, \mathbf{X}, \mathbf{Y} \mathbin{⨾} \mathbf{W}, \mathrm{dyn}_f, \mathrm{ro}_f, \mathsf{st} \rangle$ be a Moore machine. The *trace of* f *over* $\mathbf{W}$ is the Moore machine

$$\mathrm{Tr}^{\mathbf{W}}_{\mathbf{U},\mathbf{Y}} f = \langle \mathbf{U}, \mathbf{X}, \mathbf{Y}, \mathrm{dyn}_{\mathrm{Tr}\, f}, \mathrm{ro}_{\mathrm{Tr}\, f}, \mathsf{st} \rangle$$

where

$$\begin{aligned} \mathrm{dyn}_{\mathrm{Tr}\, f} : \mathbf{U} \mathbin{⨾} \mathbf{X} &\to \mathbf{X} \\ u \mathbin{⨾} x &\mapsto \mathrm{dyn}_f(u \mathbin{⨾} \mathrm{pr}_{\mathbf{W}} \mathrm{ro}(x) \mathbin{⨾} x) \end{aligned}$$

and

$$\begin{aligned} \mathrm{ro}_{\mathrm{Tr}\, f} : \mathbf{X} &\to \mathbf{Y} \\ x &\mapsto \mathrm{pr}_{\mathbf{Y}} \mathrm{ro}(x) \end{aligned}$$

Above, $\pi_{\mathbf{W}}$ is the projection $\mathbf{U} \mathbin{⨾} \mathbf{W} \to \mathbf{W}$ and $\mathrm{pr}_{\mathbf{Y}}$ is the projection $\mathbf{Y} \mathbin{⨾} \mathbf{W} \to \mathbf{Y}$.